MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, MARK 107 MOD 0

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the Mark 107 Mod 0 Impulse Cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-D-81303	Design and Evaluation of Cartridges for Stores Suspension Equipment
MIL-I-81577	Ignition Element, Electric, Mark 17 Mod 0

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-453	Inspection, Radiographic

Beneficial comments (recommendations, additions , deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1377

AMSC N/A

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

SPECIFICATIONS

Naval Sea Systems Command (Code Ident 53711)

WS 23143	Ignition Element, Electric, PVU-8/A

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

DL613468	Ignition Element, Electric, Mark 17 Mod 0
DL311AS255	Test Fixture, Cartridge, Mark 107 Mod 0
DL745AS100	Cartridge, Impulse, Mark 107 Mod 0
745AS100	Cartridge, Impulse, Mark 107 Mod 0, Assembly
745AS107	Cartridge, Impulse, Mark 107 Mod 0, Packaging Assembly
745AS108	Container, Inner, Assembly

PUBLICATIONS

Naval Sea Systems Command (Code Ident 53711)

OP 5	Ammunition and Explosives Ashore
WR-43	Preparation of Quality Assurance Provisions

Code of Federal Regulations

49 CFR 100-199	Transportation

(Copies of specifications, standards, other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Conformance to documents. The Mark 107 Mod 0 impulse cartridge covered by this specification shall be manufactured in accordance with DL745AS100 and all documents listed thereon.

3.3 Materials. All materials used in the manufacture of the cartridges shall conform strictly to the specifications referred to on the respective drawings. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Primary components. For the purposes of this specification the Mark 17 Mod 0 electric ignition element, PVU-8/A electric ignition element and the cartridge propellant charge are considered primary components.

3.5 Ignition element. The ignition elements shall be furnished from lots which were manufactured within the past 12 months. The ignition elements shall be manufactured in accordance with the drawings listed on DL613468, and shall be from lots which were accepted in accordance with the requirements of MIL-I-81577 and WS 23143 (see 4.6.1).

3.6 Propellant charge.

3.6.1 Quantity. All propellant charges shall be as specified on Drawing 745AS100. Charges shall be determined by weight rather than by volume measurement.

3.6.2 Smokeless powder, handling, and loading. The smokeless powder shall be subjected to no unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.6.3 Smokeless powder, condition. The smokeless powder used shall exhibit no evidence of decomposition (see 4.6.2)

3.7 Cartridge inspection.

3.7.1 Bridge circuit resistance. The resistance of the bridge circuit in each completely assembled cartridge, when measured electrically (see 4.6.4), shall be not less than 1.0 ohm nor greater than 1.2 ohms.

3.7.2 Leakage. Cartridge leakage rate shall be not greater than 10^{-5} cubic centimeters of air per second (see 4.6.5).

3.7.3 X-ray. Cartridges shall be free of imperfections when X-rayed (see 4.6.6).

3.8 Ballistic requirements. Cartridges shall meet the ballistic requirements specified in 3.8.1 through 3.8.3 when subjected to the test prescribed in 4.6.7.

3.8.1 Chamber pressure. The upper (U) and lower (L) limits of maximum pressure in pounds per square inch shall be 7240 and 6470, respectively.

3.8.2 Misfire. There shall be no misfires (see 4.6.7.2).

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3.8.3 Hangfire. There shall be no hangfires (see 4.6.7.3).

3.9 Change in propellant lot. Whenever a change occurs in the propellant lot for the main charge, test firings shall be conducted in accordance with 4.8.

3.10 Workmanship. Cartridges shall be free of the following defects: burrs, dents, incomplete or damaged threads, missing or damaged O-ring, perforated closure disc, or other defects which may prevent entry of the cartridge into the firing chamber, or affect the resistance of the cartridge to moisture. All cartridge marking and identification shall be verified as proper (see 4.6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed as specified in the applicable test method of 4.6.

4.4 First article inspection. First article inspection shall be performed after award of contract and prior to production by a facility designated in the contract or order (see 6.2). A first article sample acceptable for environmental and function testing shall be considered as a sample which has met the requirements of 3.2 through 3.7.3 and 3.10 of this specification. First article inspection shall be performed on sample units which have been produced using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be from the intended sources of supply.

4.4.1 Sample size. The first article sample shall consist of 94 cartridges.

4.4.2 Inspection routine. Eighty-four cartridges shall be expended in the tests listed below and 10 shall be retained for investigative purposes. Testing shall consist of the following tests as prescribed in table III of MIL-D-81303, except for temperature and humidity cycling (THC) which shall be in accordance with

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table II of MIL-D-21625: 40-foot drop test [(6) nonfiring test]; 15 "g" shock (12); THC (18); vibration (12); −65°F temperature (12); +200°F temperature (12); +70°F temperature (12). The number in parentheses denotes the number of cartridges allocated for each test. After exposure to the specified environmental treatment, the cartridges allocated for the 15 "g" shock, THC, vibration, and normal temperature shall be conditioned for a minimum of 6 hours at 70° ± 5°F and fired within 5 minutes after removal from conditioning. The cartridges allocated for the low temperature and high temperature tests shall be conditioned at −65° ± 5°F and 200° ± 5°F, respectively, for not less than 6 nor more than 24 hours and fired within 3 minutes after removal from conditioning. All first article cartridge firings shall be conducted in a test fixture in accordance with the requirements of 4.6.7.1. All cartridges shall fire and produce results within the limits specified in 3.8.1 and 3.8.3, except that the upper (U) and lower (L) limits of maximum pressure in pounds per square inch for cartridges fired at −65°F and 200°F shall be U = 7630 and L = 6220. The pressure limits of 3.8.1 apply for all cartridges fired at 70°F.

4.4.3 Failure. Failure of any cartridge shall be cause for rejection of the first article sample.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the tests specified in table I.

TABLE I. Quality conformance tests and examinations.

Item	Nature of test	Requirement para.	Test para.	Classification (WR-43)
Primary component	Single lot verification	3.4	4.5.1	(M101)
Ignition element	Age and acceptance verification	3.5	4.6.1	(C1)
Propellant charge	Type and weight	3.6.1	-	(M102)
Smokeless powder	Condition	3.6.3	4.6.2	(M103)
Cartridge	Visual examination	3.7.4	4.6.3	(M104)
Cartridge	Bridge circuit resistance	3.7.1	4.6.4	(C2)
Cartridge	Leakage	3.7.2	4.6.5	(C3)
Cartridge	X-ray	3.7.3	4.6.6	(C4)
Cartridge	Ballistics	3.8	4.6.7	(C5)
Cartridge	Ballistics (change in propellant lot)	3.9	4.8	(C6)
Cartridge	Preservation-packaging, packing and marking	Section 5	4.6.3	(M105)

4.5.1 Lot size. Unless otherwise specified (see 6.2), the production lot size shall be not less than 1301 and not greater than 3200. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot.

4.5.2 Sampling. Unless otherwise specified (see 6.2), sampling shall be in accordance with table II.

TABLE II. Quality conformance sample size.

Lot size	Sample size	Retained sample
66 - 110	15	10
111 - 180	15	10
181 - 300	20	10
301 - 500	25	10
501 - 800	30	10
801 - 1300	35	10
1301 - 3200	40	10
3201 - 8000	50	10
8001 - 22000	75	10

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4.5.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.6 Tests.

4.6.1 Ignition element verification. Verification shall be provided that the ignition elements used were certified and accepted in accordance with the requirements of 3.5.

4.6.2 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable paragraphs of OP 5. Evidence of decomposition shall be cause for rejection.

4.6.3 Visual examination. All cartridges shall be visually examined. Cartridges having any of the defects of 3.7.4 shall be cause for rejection. The preservation-packaging, packing, and marking of the cartridges shall be examined for conformance to section 5.

4.6.4 Bridge circuit resistance. Conformance with the bridge circuit resistance of 3.7.1 shall be ascertained on all completely assembled cartridges by means of a test circuit which limits the bridge circuit current to not greater than 25 milliamperes. A safety chamber with an interlock switch shall be used to protect the operator during the application of current. A suitable instrument which is known to be accurate to within 1 percent in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units which fail to meet the requirements of 3.7.1 shall be cause for rejection. The results of bridge circuit resistance measurements shall not be affected by resistance of the test circuit and connector.

4.6.5 Leakage. Each cartridge in the first article or production sample shall be leak tested in a dry gas leak tester. Cartridges which exhibit a leak rate greater than 10^{-5} cc per second of air at a pressure differential of 1 ± 0.1 atmosphere shall be considered defective. Cartridges of the first article sample which exhibit a leakage defect shall be cause for rejection. If the number of cartridges of the production lot sample exhibiting leak defects exceeds an AQL of 2.5 percent defective from table II-A of MIL-STD-105, then a 100 percent leak test of the lot being tested shall be required. Cartridges which exhibit a leak rate in excess of that specified above shall be cause for rejection (see 6.5).

4.6.6 Radiographic examination. All cartridges, in the first article and production test samples, shall be radiographically examined in accordance with MIL-STD-453. Cartridges having any observable imperfections in assembly shall be cause for rejection. The cartridges shall be positioned on their sides for the most revealing exposure. All cartridges shall be identified with serial numbers prior to examination. The cartridge serial numbers shall be in consecutive order beginning with the number 001 in each production lot. The cartridges shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The radiograph identification shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. If any assembly or loading defects are noted in the production lot samples, the contractor must X-ray the represented lot 100 percent and submit the X-rays to the cartridge design activity for review. Discontinuities in serial numbers shall be identified on the X-ray. Defective cartridges found by the vendor shall be clearly identified on the X-ray and those defectives removed from the production lot. The serial numbers shall be pen or ink stamped on the cartridges with approximately 1/8-inch-high characters. The serial numbers shall be located apart from the other cartridge markings.

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4.6.7 Ballistic tests. The number of cartridges selected from each production lot for the ballistic test shall be in accordance with table II. The cartridges shall be conditioned for not less than 6 hours at 70° ± 5°F and fired in a closed bomb test fixture manufactured in accordance with the documents listed on DL311AS255. Each cartridge shall be fired within 5 minutes after removal from conditioning. A record of pressure versus time and of the elapsed time between actuation of the firing mechanism and start of pressure rise shall be taken with suitable instrumentation for each firing. The voltage source used for the firing shall be 28 ± 2 volts dc, and the firing circuit shall be regulated to produce a firing current of 10 ± 0.5 amperes. In addition to the sample of cartridges for the above test, 10 additional cartridges shall be furnished from each production lot for investigative purposes. Acceptance criteria (regardless of lot size) shall be: accept zero defects and reject if one or more defects.

4.6.7.1 Criteria of acceptability. The upper (U) and lower (L) limits of maximum pressure shall meet the requirements of 3.8.1. A failure of any cartridge to meet this requirement shall be cause for rejection of the lot represented.

4.6.7.2 Misfire. The failure of any cartridge to fire shall be cause for rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.6.7.3 Hangfire. One or more hangfires (see 6.4.5) shall be cause for rejection of the lot represented.

4.7 Retest. There shall be no retests.

4.7.1 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridge, original test results should be discarded and that portion of the test reconducted.

4.8 Change in propellant lot. Whenever a change in propellant lot for the main charge occurs, the charge weight shall be verified by firing 36 cartridges which have been temperature conditioned as follows: 12 at −65° ± 5°F, 12 at 70° ± 5°F, and 12 at 200° ± 5°F. The temperature conditioning and firing of these cartridges and the firing results shall be the same as the cartridge allocated for the low, high, and normal temperature firing tests specified in 4.4.2.

5. PACKAGING

5.1 Preservation and packaging. For definitions of packaging levels, see 6.4.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. Four cartridges shall be packaged in an inner container in accordance with Drawing 745AS108.

5.1.2 Level B. Not applicable.

5.1.3 Level C. When directed by the procuring activity, those cartridges which are intended for expenditure in first article sample testing and production lot acceptance testing shall be packaged in accordance with method III of MIL-P-116 and with 49 CFR 171-178.

5.2 Packing. For definitions of packing levels, see 6.4.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. Twelve inner containers as described in 5.1.1 shall be packed in accordance with Drawing 745AS107.

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5.2.2 **Level B.** Not applicable.

5.2.3 **Level C.** When directed by the procuring activity, cartridges intended for expenditure in first article testing and for production lot acceptance tests and are packaged in accordance with 5.1.3 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with 49 CFR 171-178.

5.3 **Marking.**

5.3.1 **Special marking.** Marking of exterior containers shall be in accordance with 49 CFR 171-178.

5.3.2 **Normal marking.** Unless otherwise specified in the contract or order, the marking information on unit packages and shipping containers shall be as specified on Drawings 745AS108 and 745AS107, respectively. The specified marking information shall be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

6. NOTES

6.1 **Intended use.** The Mark 107 Mod 0 impulse cartridge is intended to provide power to stores from the F-14 aircraft.

6.2 **Ordering data.** Acquisition documents should specify the following:

a. Title, number and date of this specification
b. Whether first article inspection is required, and if so, specify the test activity
c. Date and place of delivery
d. Inspection conditions, if other than as specified (see 4.3)
e. That revision "D" of MIL-D-21625 be used (see 4.4.2)
f. Lot size, if other than as specified (see 4.5.1)
g. Sampling, if other than as specified (see 4.5.2)
h. Marking, if other than as specified (see 5.3.2)
i. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DoD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP5 are applicable.

6.3 **First article.** When a first article inspection is required, the item should be a first article sample. The first article should consist of 94 cartridges. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 **Definitions.** For the purpose of this specification, the following definitions apply.

6.4.1 **Level A packaging.** The degree of preservation-packaging which will afford adequate protection against corrosion, deterioration and physical damage during handling, shipment, indeterminate storage and world-wide redistribution.

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6.4.2 **Level C packaging.** The degree of preservation-packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

6.4.3 **Level A packing.** The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.4 **Level C packing.** The degree of packing which will afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

6.4.5 **Hangfire.** A hangfire is defined as a firing in which the elapsed time between actuation of the firing mechanism and the start of pressure rise exceeds 0.03 second or time from start of pressure to peak pressure exceeds 0.05 second.

6.5 **Advisory leak test.** Indian Head leak tests cartridges in accordance with "Cartridge Leak Test Procedures at the Naval Ordnance Station, Indian Head, Md."

6.6 **Changes from previous issue.** Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-81984A (AS)	CARTRIDGE, IMPULSE, MARK 107 MOD 0

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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